Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/24/2007
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Project Title:	THERMAL TRANSPIRATION BASED MICOSCALE PROPULSION AND POWER GENERATION		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/23/2003	**End Date:**	10/01/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GA29G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The primary goal of this project is to develop a material system for use in a catalytic-combustion driven thermal transpiration-based pump which will be applicable to space or airbreathing vehicles. This pump will have no moving parts, no pressurization fluids, very low mass, and will integrate pressurization and thrust generation. The material system that needs to be designed for this pump must be very porous with pore sizes comparable to the mean free path of air molecules, must be an efficient catalyst support, must have high mechanical stability, and must have low thermal conductivity. The first objective of this project is to design a nanoporous/mesoporous transpiration membrane that can be fabricated into a free-standing monolith disc and can be tested in a single stage thermal transpiration pump. Our second objective is to increase the pumping efficiency of the system by creating an controlled aerogel microstructure that is ideal for transpiration flow. The third objective involves eliminating the need for electrical heating as the heating element for		

	thermal transpiration. Instead, the heat source to initiate thermal transpiration pumping in our proposed system will be generated by catalytic combustion. We plan to deposit platinum nanoparticles onto one surface of the transpiration membrane to create a catalyst support that will not reduce the thermal transpiration performance of the membrane material. Our project will aid in the development of a very efficient motor-less pump that will have no pressurization fluids, very low mass, and will integrate pressurization and pump generation. The pump will also be adaptable into a microscale power generation device for use in a variety of mobile applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A main goal in this project is to develop a very efficient pump/power generation device as an alternative to current power generation devices. Our proposed system can be designed so that the only products of the power generation system are water and carbon dioxide. Another main goal is to produce novel materials and architechtures that demonstrate innovative engineering to improve device performance. A key technology that has already been demonstrated for this project is the adaptation of an ordered three-dimensional macroporous structure aimed at improving gasous flow in thermal transpiration, catalytic, and thermoelectric systems.
Task Progress:	We had three major aims for our research for this year which focused on developing enhanced materials for our thermal transpiration device. Our first goal was to develop a stronger and less brittle silica-based aerogel by incorporating organic ligands in the silica matrix to allow for greater mechanical strength without compromising the thermal properties of the transpiration membrane. We successfully incorporated a methyl-terminated silica precursor into the aerogel matrix which increased the mechanical integrity of the aerogel membrane. Not only will this allow for thinner membranes to be fabricated (which will increase the output gas flow of the device) the addition of the methyl group also changes the surface chemistry of the material. The traditional aerogel matrix is hydrophilic and will degrade if water is continuously adsorbed and desorbed from the material. The organically modified aerogel helps to prevent this type of water adsorption by increasing the hydrophobicity of the material. The resulting gels remain transparent and very porous with theoretically comparable thermal properties as traditional aerogels. Our second goal was to develop a different morphology for our platinum catalytic particles to increase gas flow kinetics and allow a greater contact area with the aerogel membrane. We have previously demonstrated the deposition of platinum catalytic particles onto the aerogel substrate in order to serve as a catalyst for hydrocarbon combustion and provide a heating source for the transpiration membrane. However, to improve the performance of the catalyst we have successfully fabricated a porous platinum powder that can be deposited onto the surface of the aerogel that will improve the kinetics of the fuel that enters the system. The porous structure allows for quick venting of reacted products (mainly water and carbon dioxide) which enables new fuel to enter the system at a much faster rate. The macroporous platinum powders were fabricated using a polystyrene bead template approach. Using a platinum salt that was reduced in ethylene glycol, we synthesized platinum nanoparticles which were then washed in an alcohol-based solution. The nanoparticles were then combined with the polystyrene beads and were sintered to allow for the growth of the particles. Also, we measured the surface area of the platinum powders to be 111.52 m^2/g which will provide a large number of catalytically active sites for the combustion of the incoming fuel. Our last goal was to begin synthesis of thermoelectric materials that can be incorporated into the structure as an energy harvesting method from the heat generated from the catalytic reaction. As a proof of concept system, we used a ceria-zirconia system to show the feasibility of making a thermoelectric macroporous structure which can be directly applied to a thermoelectric material system (e.g. bismuth). We synthesized a nanoparticulate sol of ceria-zirconia and used an ordered array of polystyrene beads as a macroporous template. The system was sintered to thermally decompose the polystyrene beads and also to allow some crystal growth of the ceria. The resulting structure is a very ordered and three-dimensional network of pores that extended throughout the material. The same type of synthesis route could be easily applied to a thermoelectric system such as bismuth and because of the macroporous nature of the material, we can now incorporate this layer into our device without greatly reducing the gas flow through the membrane.
Bibliography Type:	Description: (Last Updated: 02/11/2020)
Articles in Peer-reviewed Journals	Umeda GA, Chueh WC, Noailles L, Haile SM, Dunn BS. "Inverse opal ceria–zirconia: architectural engineering for heterogeneous catalysis." Energy & Environmental Science. 2008;1:484-6. http://dx.doi.org/10.1039/B810641J , Jan-2008